

```
#include <stdio.h>
```

```
Int main()
```

```
{
```

```
    Int pages[30], frame[10];
```

```
    Int n, f, l, j, k, pos, far, next, hit, fault = 0;
```

```
    Printf("Enter number of pages: ");
```

```
    Scanf("%d", &n);
```

```
    Printf("Enter reference string:\n");
```

```
    For(l = 0; l < n; i++)
```

```
        Scanf("%d", &pages[i]);
```

```
    Printf("Enter number of frames: ");
```

```
    Scanf("%d", &f);
```

```
    For(l = 0; l < f; i++)
```

```
        Frame[i] = -1;
```

```
    Printf("\nPage\t");
```

```
    For(l = 0; l < f; i++)
```

```
        Printf("F%d\t", l + 1);
```

```
    Printf("Status\n");
```

```
    For(l = 0; l < n; i++)
```

```
{  
    Hit = 0;  
  
    For(j = 0; j < f; j++)  
    {  
        If(frame[j] == pages[i])  
        {  
            Hit = 1;  
            Break;  
        }  
    }  
}
```

```
If(!hit)
```

```
{  
    Pos = -1;
```

```
    For(j = 0; j < f; j++)
```

```
    {  
        If(frame[j] == -1)
```

```
        {  
            Pos = j;  
            Break;
```

```
        }  
    }  
}
```

```
If(pos == -1)
```

```

{
    Far = -1;

    For(j = 0; j < f; j++)
    {
        Next = n;

        For(k = l + 1; k < n; k++)
        {
            If(frame[j] == pages[k])
            {
                Next = k;
                Break;
            }
        }

        If(next > far)
        {
            Far = next;
            Pos = j;
        }
    }

    Frame[pos] = pages[i];
    Fault++;
}

```

```
}
```

```
Printf("%d\t", pages[i]);
```

```
For(j = 0; j < f; j++)
```

```
{
```

```
    If(frame[j] == -1)
```

```
        Printf("-\t");
```

```
    Else
```

```
        Printf("%d\t", frame[j]);
```

```
}
```

```
If(hit)
```

```
    Printf("Hit\n");
```

```
Else
```

```
    Printf("Fault\n");
```

```
}
```

```
Printf("\nTotal Page Faults = %d\n", fault);
```

```
Return 0;
```

```
}
```